

Linear Programming Course Summary

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1 Key Terms

1.1 linear programming fundamentals

- **Slack Variable:** *A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. The slack variable represents the difference between the left-hand side and the right-hand side of the inequality.*

1.2 solution space analysis

- **Feasible Region:** *The set of all points that satisfy all the constraints of a linear programming problem.*
- **Vertex:** *A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.*

1.3 simplex method algorithms

- **Basic Variable:** *In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. Basic variables are set to values that satisfy the constraints.*
- **Non-basic Variable:** *In the Simplex method, a variable that is set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.*
- **Pivot Operation:** *The process of exchanging a basic variable with a non-basic variable in the Simplex method, leading to a new dictionary and a movement to a different vertex in the feasible region.*
- **Dictionary:** *A representation of a linear programming problem in a tabular format, where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.*
- **Auxiliary Problem:** *A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.*

2 Problem Types

2.1 linear programming fundamentals

- **Finding an Initial Feasible Solution:** *The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.*
- **Using Dictionaries to Represent Solutions:** *The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.*

2.2 advanced linear programming

- **Finding an Initial Feasible Solution:** *The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.*
- **Handling Infeasible Origins:** *The origin $(0, 0)$ may not satisfy all constraints in a linear programming problem. Techniques like introducing an auxiliary problem are used to find a feasible starting point.*
- **Optimizing in Higher Dimensions:** *The Simplex method is extended to linear programming problems with more than two variables, requiring iterative pivoting operations to find the optimal solution.*
- **Using Dictionaries to Represent Solutions:** *The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.*

3 Algorithm Solutions

3.1 linear optimization techniques

- **Simplex Method:** *Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.*

3.2 problem formulation and solution strategies

- **Auxiliary Problem:** *When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem to find a feasible solution for the original problem. If the optimal solution has $x_0 = 0$, then a feasible solution to the original problem is found.*